# Yuan (Merlin) Chang

Email: merlin.y.chang@gmail.com
Github: github.com/itis2010me
Mobile: +1 (530) 760 6690
Website: itis2010me.github.io/itis2010me

## Experience

- **AWS Annapurna Labs - Neuron Automated Reasoning Group** — Seattle, WA
  *Software Development Engineer II (Compiler Engineer)* — *Oct 2026 – Present*
  - Building a formally verified language stack for writing highly optimized vector engine kernels.
  - Building a formally verified assembler for the Trainium/Inferentia instruction set architecture (ISA).
  - Building simulation of AWS Trainium machine learning accelerator hardware.
- **Amazon Web Services (AWS) - Automated Reasoning Group** — Seattle, WA
  *Software Development Engineer II* — *Jul 2026 – Oct 2026*
  *Software Development Engineer I* — *Feb 2025 – Jul 2026*
  - Built an AI agent tool call evaluation system using formal policies such as Dogwood to improve agent safety.
  - Raised the compute fleet scaling ceiling by 20x (3k to 60k) and cut per-unit scale-up time by 60x through a backend migration of a highly concurrent, distributed service to Firecracker microVMs.
  - Lowered the barrier for internal builders and scientists to adopt formal reasoning engines (SAT/SMT, Lean4) by designing, implementing and shipping remote stateful MCP servers.
  - Eliminated outages from partial availability-zone dependency failures with Elastic Load Balancing over deep health checks, and the top recurring source of on-call pages with idempotent, order-tolerant ECS event handling.
  - Raised the team's operational bar with a fault injection framework, regular game days, and ordered, automated deployment sequencing that removed cross-pipeline dependencies.
  - Mentored junior engineers and interns through design and code review, scoping and guiding an intern project that converted to a full-time offer.
- **Amazon Web Services (AWS) - High Performance Computing** — Seattle, WA
  *Software Development Engineer I* — *Apr 2024 – Feb 2025*
  - Shipped AWS Parallel Computing Service (PCS), a public managed HPC service, as part of the launch team, building across the AWS Console frontend, the backend service APIs, and the workflows that provisioned customer compute clusters.
  *Software Development Engineer Intern* — *Jun 2023 – Sep 2023*
  - Designed internal RESTful APIs and orchestration that collapsed multi-step cluster configuration and job scheduling into single API invocations.
  - Shipped a public AWS CLI command exposing the managed HPC service directly to enterprise customers.
- **UC Davis Applied Mathematics Research Group** — Davis, CA
  *Research Assistant & Software Engineer* — *Jun 2021 – Jun 2022*
  - Developed an automated reasoning pipeline that computed nearly 500 new Ramsey constants.
  - Reduced the search space by 6x through symmetry breaking and optimization techniques in preprocessing.
  - Achieved up to 10x improvement in SAT clause generation through computation optimizations.

## Technical Skills

- **Languages** — Java, TypeScript, Bash, Rust, Python
- **Distributed Systems** — REST APIs, NoSQL, Firecracker microVMs, containerization (Docker), image-based deployment
- **Cloud & Build** — AWS (Lambda, ECS, SQS, DynamoDB, ElastiCache, S3, ELB), build systems, CI/CD tooling, Git
- **Agentic AI** — MCP servers, agent skills, steering docs, Claude Code, Kiro

## Education

- **University of California, San Diego** — La Jolla, CA
  *Master of Science - Computer Science and Engineering* — *2022 – 2024*
- **University of California, Davis** — Davis, CA
  *Bachelor of Science - Computer Science, Minor in Mathematics; Graduated with Honors* — *2018 – 2022*

## Publications

- **Rado Numbers and SAT Computations (with J. A. De Loera and W. J. Wesley).**
  Proceedings of the 47th International Symposium on Symbolic and Algebraic Computation (ISSAC 2022). Pages 333-342, available online at https://dl.acm.org/doi/10.1145/3476446.3535494.

## Projects

- **Minimization of CA Wildfire Impact (Convex Optimization, CVXPY)**: Prioritized fire suppression across 58 CA counties under budget constraints via the Lagrangian dual, KKT conditions, and a near-integral convex relaxation.